The energy that becomes thermal

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for mechanical energy changes through heating and deformation.

SUCCESS CRITERIA - I can...

1. Track system boundaries in an energy balance.
2. Distinguish thermal energy from temperature.
3. Use energy differences to quantify a transfer.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “thermal energy” and “system” to explain your model or result. Sentence starters are optional.

thermal energy: internal energy associated with microscopic motion/interactions

system: the object or objects chosen for analysis

conserved: unchanged in total under the stated conditions

01 Bring back an earlier idea

A spring has k=300 N/m and stores 0.375 J.
Find displacement magnitude and force magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $E_{mechanical,\ i} = E_{mechanical,\ f} + \Delta E_{thermal}$.
No other transfers in this model; include contacting materials in the system. Friction transfers mechanical energy to internal energy.

Worked reasoning: A 2 N friction force opposes a sliding box over 5 m. It does -10 J of work; in a suitable box-floor system this appears as a 10 J thermal-energy increase.

Your turn: Draw the force, displacement and energy transfer.
Name the system for your energy account.

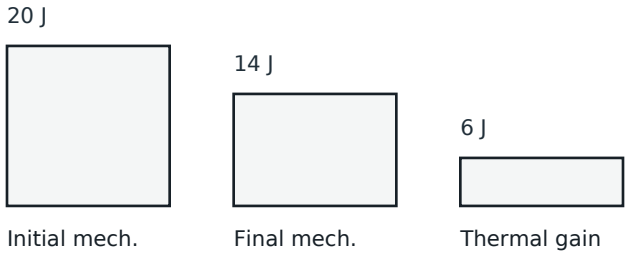
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

A cart begins with 20 J mechanical energy and ends with 14 J. The closed accounting includes 6 J thermal increase. Does it balance?

Sentence starter: I notice ... . This supports ... because ... .



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Solve and justify

A 2 kg object slows from 5 to 2 m/s on a horizontal surface.
Find mechanical energy loss. If all becomes internal energy of object+surface, find that increase.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Does the same energy input always produce the same temperature rise in two different objects?
Name quantities you would need.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

Net force rises linearly from zero to 12 N over 0.4 m.
Find work and final speed for a 0.4 kg cart starting at rest.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A .50 kg object at 6 m/s stopsthroughfriction.
Find energy transformed. If stopping distance is 1.5 m, find average opposing force.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 3 kg box slides down a 4 m ramp inclined 30° with friction magnitude 5 N. Starting at rest and usingg=10, find final speed.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3